AP PHYSICS C: ELECTRICITY & MAGNETISM · THE UNIVERSAL STUDY GUIDE

Calculus-based — targeting a 3, 4, or 5

Tier badges: [3] foundational · [4] adds for 4+ · [5] adds for 5

Targeting a 4 → study [3] AND [4] items. Targeting a 5 → study all three tiers.

PAGE 1 | MATH OF 3/4/5 · 5 UNITS · GAUSS · AMPÈRE · FARADAY

About this exam. AP Physics C: E&M is the calculus-based companion to Mechanics — same exam structure (35 MCQ + 3 FRQ in 90 min), but the toolkit is **integrals over charge distributions, line integrals, surface integrals, and ODEs** for RC/RL/LC circuits. **Symmetry-based applications of Gauss's Law and Ampère's Law** are the signature FRQ challenge.

THE MATH OF YOUR TARGET

Target	MCQ (35q, 50%)	FRQ (3q, 50%)	Approx %	Tier
3 (PASS)	**~16 / 35** (~46%)	**~14/45 raw**	~40-44%	[3]
4	**~22 / 35** (~63%)	**~22/45 raw**	~52-58%	[3]+[4]
5	**~28 / 35** (~80%)	**~32/45 raw**	~70-75%	All

Time: 45 min for 35 MCQ (~1:17 each), 45 min for 3 FRQs (~15 min each). Calculator allowed throughout. Reference table provided.

THE 5 UNITS — by exam weight

#	Unit	Weight	Focus
1	Electric Charge & Field	26–34%	Coulomb's law, fields from continuous distributions, **Gauss's Law**
2	Electric Potential & Potential Energy	14–17%	V = ∫E·dl, potential of distributions, equipotentials
3	Conductors, Capacitors, Dielectrics	14–17%	Conductor properties, capacitance, energy stored
4	Electric Circuits	17–23%	Kirchhoff, **RC circuits with ODEs**, time constants
5	Magnetic Fields & Electromagnetic Induction	17–23%	**Biot-Savart**, **Ampère's Law**, Faraday/Lenz, RL circuits, LC oscillation

> **KEY INSIGHT** **[5] Unit 1 alone is 26-34% — the largest single unit on any AP exam.** Master Gauss's Law for spherical, cylindrical, and planar symmetry. Most FRQs hinge on it.

THE 6 CALCULUS DEFINITIONS YOU MUST KNOW [3]/[4]

```
E from continuous charge:    E = ∫ k·dq / r²       (vector — set up by component)

V from continuous charge:    V = ∫ k·dq / r        (scalar — easier!)

E from V:    E = -dV/dr     (or E_x = -∂V/∂x in multiple dimensions)

GAUSS'S LAW:    ∮ E·dA = Q_enc / ε₀

AMPÈRE'S LAW:    ∮ B·dl = μ₀ I_enc

FARADAY'S LAW:  EMF = -dΦ_B/dt
  where Φ_B = ∫ B·dA
```

PAGE 2 | UNIT 1: ELECTRIC FIELD · INTEGRATION · GAUSS'S LAW

Coulomb's Law (point charges) [3]/[4]

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POINT CHARGES (vector form):    F = (kq₁q₂/r²) r̂

E from a POINT CHARGE:    E = (kQ/r²) r̂        (away from + Q)

Constants:    k = 1/(4πε₀) = 8.99 × 10⁹ N·m²/C²
              ε₀ = 8.85 × 10⁻¹² C²/(N·m²)
```

E from Continuous Charge Distributions [4]/[5]

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GENERAL APPROACH:
 1. Choose dq element (small chunk of charge)
 2. Express dq in terms of integration variable:
      Linear charge density λ:  dq = λ dl  (rod, wire)
      Surface density σ:        dq = σ dA  (disk, plate)
      Volume density ρ:         dq = ρ dV  (sphere, cylinder)
 3. Write dE from dq:  dE = k·dq / r²
 4. Identify symmetry — what components CANCEL?  (only integrate the surviving component)
 5. Integrate over the distribution.

EXAMPLE — point on axis of uniformly charged ring (radius R, total Q, distance z from center):
  By symmetry, only z-component survives.
  dE_z = k·dq / (z² + R²) · cos θ  where cos θ = z / √(z² + R²)
  E_z = kQz / (z² + R²)^(3/2)
```

Gauss's Law — the symmetry shortcut [3]/[4]/[5]

```
∮ E · dA = Q_enclosed / ε₀

USE WHEN: charge distribution has SPHERICAL, CYLINDRICAL, or PLANAR symmetry.

STRATEGY:
 1. Identify symmetry → choose Gaussian surface where E is constant or perpendicular
 2. Pull E out of the integral:  ∮ E·dA = E·A
 3. Compute Q_enclosed (depends on Gaussian surface size)
 4. Solve for E

CLASSIC RESULTS:
  Point charge / outside sphere of charge:  E = kQ/r²
  Inside uniformly charged solid sphere:    E = kQr/R³  (linear in r!)
  Inside conducting sphere:                 E = 0
  Infinite line of charge (linear λ):       E = λ/(2πε₀r)
  Infinite plane of charge (σ):             E = σ/(2ε₀)
```

KEY INSIGHT **[5] Why E = 0 inside a conductor.** Free charges rearrange until they cancel any internal field. This is electrostatic equilibrium. Charge sits ENTIRELY on the surface. Use this when applying Gauss's Law to conductors.

Continuous distribution master table [4]/[5]

Distribution	Field formula
Infinite line, linear λ	$E = \lambda/(2\pi\varepsilon_0 r)$ — falls as $1/r$
Infinite plane, σ	$E = \sigma/(2\varepsilon_0)$ — UNIFORM (independent of distance)
Two parallel plates (capacitor)	$E = \sigma/\varepsilon_0$ between plates, ~0 outside
Solid sphere, total Q, uniform	Outside ($r > R$): $E = kQ/r^2$. Inside: $E = kQr/R^3$
Spherical shell, total Q	Outside: $E = kQ/r^2$. Inside: $E = 0$

PAGE 3 | UNITS 2-3: POTENTIAL · CAPACITORS · DIELECTRICS

Electric Potential V [3]/[4]

```
POINT CHARGE:    V = kQ/r       (taking V → 0 at infinity)

CONTINUOUS DISTRIBUTION:    V = ∫ k·dq / r
  SCALAR — much easier than E (no vector decomposition needed)

POTENTIAL DIFFERENCE:    ΔV = V_b − V_a = −∫_a^b E·dl

WORK to move charge q from a to b:    W = qΔV = q(V_b − V_a)

PE of charge configuration (point charges):    U = ½ Σ (kq_iq_j / r_ij)
```

E from V [4]/[5]

```
In 1-D:    E_x = −dV/dx

In 3-D:    E = −∇V = −(∂V/∂x x̂ + ∂V/∂y ŷ + ∂V/∂z ẑ)

PHYSICAL MEANING: E points in direction of STEEPEST DECREASE of V
                   (downhill for positive charge).

Equipotential surfaces: lines of constant V.  E ⊥ equipotential everywhere.
```

KEY INSIGHT **[5] Compute V first, then E from V.** When asked for E from a continuous distribution, computing V (scalar integral) and then taking $E = -dV/dx$ is often EASIER than direct E integration (vector). Especially on axis of symmetry.

Conductors in Electrostatic Equilibrium [3]/[4]

- **[3] E = 0 inside the conductor** (charges rearrange until cancellation).
- **[3] All charge on the SURFACE.**
- **[3] Surface is an equipotential** (V = constant on conductor).
- **[4] E just outside conductor**: perpendicular to surface, magnitude = σ/ε_0.
- **[4] Charge concentrates at points/sharp corners** (lightning rod principle).

Capacitors [3]/[4]

```
Q = CV          C = capacitance (Farads)

PARALLEL PLATE:    C = ε₀ A / d    (vacuum)
                   C = κ ε₀ A / d  (with dielectric κ)

STORED ENERGY:    U = ½ CV² = ½ Q²/C = ½ QV
ENERGY DENSITY:   u = ½ ε₀ E²  (energy per volume in field)

COMBINING (OPPOSITE of resistors):
  Series:    1/C_eq = Σ(1/C_i)        (less than smallest)
  Parallel:  C_eq = ΣC_i               (just add)
```

Dielectrics [4]/[5]

- **[4] Insulator inserted between capacitor plates.** Dielectric constant $\kappa > 1$.
- **[4] With dielectric**: capacitance $C \rightarrow \kappa C$, energy stored... depends on whether battery is connected.
- **[5] Battery DISCONNECTED**: Q stays constant. $V = Q/C$ drops. $U = Q^2/(2C)$ drops.
- **[5] Battery CONNECTED**: V stays constant. $Q = CV$ increases. $U = \frac{1}{2}CV^2$ increases.

PAGE 4 | UNIT 4: CIRCUITS — KIRCHHOFF · RC ODEs

Kirchhoff's Laws [3]/[4]

KCL (junction): $\Sigma I_{in} = \Sigma I_{out}$
KVL (loop): $\Sigma \Delta V = 0$ around any closed loop

PATTERN for multi-loop circuits:

1. Label currents I_1, I_2, ... in each branch (assume direction).
2. Apply KCL at junctions.
3. Apply KVL to each independent loop.
4. Solve the system. Negative answer = current actually flows opposite way.

RC Circuit ODEs [4]/[5]

CHARGING (battery V_0, resistor R, capacitor C, switch closes at t=0):
KVL: $V_0 - IR - Q/C = 0$
Substitute $I = dQ/dt$: $R(dQ/dt) + Q/C = V_0$
SOLUTION: $Q(t) = CV_0(1 - e^{-t/\tau})$ $\tau = RC$
$I(t) = (V_0/R) e^{-t/\tau}$

DISCHARGING (charged C, R, switch closes at t=0):
KVL: $Q/C - IR = 0$ with $I = -dQ/dt$
SOLUTION: $Q(t) = Q_0 e^{-t/\tau}$
$I(t) = (Q_0/RC) e^{-t/\tau}$

TIME CONSTANT $\tau = RC$. After τ, charged to ~63% (or discharged to ~37%).
After 5τ, essentially complete (~99%).

Initial vs Steady State [3]/[4]

Element	At t = 0 (just after switch)	At t → ∞
Capacitor	Acts like WIRE (Q = 0, V_cap = 0)	Acts like OPEN (I = 0)
Inductor	Acts like OPEN (I = 0)	Acts like WIRE (V_L = 0)

KEY INSIGHT **[4] Use these limits to simplify circuits dramatically.** At t = 0 with capacitor, redraw with cap as wire → solve for initial currents. At t = ∞, redraw with cap as open → solve for steady-state currents and voltages.

Power + Energy in Circuits [3]/[4]

$P = IV = I^2R = V^2/R$ (instantaneous, in resistors)

Capacitor: $U_{cap} = \frac{1}{2}CV^2$, power stored: $P = dU/dt$
Battery: $P_{supplied} = \varepsilon I$

ENERGY DISSIPATED in resistor over discharge:
$E_{dissipated} = \int_0^\infty I^2R\,dt = \frac{1}{2}CV_0^2$ (= initial cap energy)

PAGE 5 | UNIT 5 PT 1: MAGNETIC FIELDS — BIOT-SAVART · AMPÈRE

Magnetic Force on Charges + Wires [3]/[4]

ON MOVING CHARGE: $\vec{F} = q\vec{v} \times \vec{B}$ (vector cross product)
$|F| = qvB \sin\theta$ direction: right-hand rule

ON CURRENT-CARRYING WIRE: $\vec{F} = I\vec{L} \times \vec{B}$
$|F| = BIL \sin\theta$

CHARGED PARTICLE in uniform B perpendicular to v:
Circular motion. $r = mv/(qB)$ $T = 2\pi m/(qB)$
KE unchanged (B does NO WORK on charges).

Biot-Savart Law [4]/[5]

$$dB = (\mu_0/4\pi) \cdot (I\, dl \times \hat{r}) / r^2$$

$$\mu_0/(4\pi) = 10^{-7}\ \mathrm{T\cdot m/A}$$

USE WHEN: finding B from a wire of arbitrary shape (no high symmetry).

CLASSIC RESULTS:
Infinite straight wire: $B = \mu_0 I/(2\pi r)$
On axis of circular loop (radius R, distance z):
$B = \mu_0 I R^2 / (2(z^2 + R^2)^{3/2})$
At center of circular loop: $B = \mu_0 I/(2R)$

Ampère's Law — symmetry shortcut [4]/[5]

$$\oint B \cdot dl = \mu_0 I_{enclosed}$$

USE WHEN: high symmetry (long wires, solenoids, toroids).

STRATEGY:
1. Choose Amperian loop where B is constant or perpendicular to dl.
2. $\oint B\cdot dl = B \cdot L$ (where L is length of loop where B is along dl).
3. I_{enc} = current piercing the loop.
4. Solve for B.

CLASSIC RESULTS:
Infinite straight wire (Ampère): $B = \mu_0 I/(2\pi r)$ (matches Biot-Savart)
Long solenoid (n turns/length): $B = \mu_0 n I$ (uniform inside, ~0 outside)
Toroid: $B = \mu_0 N I / (2\pi r)$ (inside)

KEY INSIGHT **[5] Gauss/Ampère analogy.** Both work via SYMMETRY. Gauss for E from charge distributions (spherical/cylindrical/planar). Ampère for B from current distributions (cylindrical/solenoidal/toroidal). Same setup pattern: pick surface/loop, exploit symmetry, set up integral, solve.

PAGE 6 | UNIT 5 PT 2: INDUCTION · RL CIRCUITS · LC OSCILLATION

Magnetic Flux + Faraday's Law [3]/[4]

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MAGNETIC FLUX:    Φ_B = ∫ B · dA  =  B·A·cos θ   (uniform B)

FARADAY'S LAW:    EMF = −dΦ_B/dt
  The EMF drives a current that opposes the CHANGE in flux (Lenz).

HOW TO CHANGE Φ_B:
  Change B (change source: stronger/weaker, or move magnet)
  Change A (loop expanding/shrinking)
  Change θ (rotate loop in B field — this is how generators work)
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Lenz's Law — direction of induced current [3]/[4]

[3] Induced current creates a B field that OPPOSES the change in flux.

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EXAMPLE: bar magnet (N-pole down) approaching loop from above.
  Flux DOWN through loop is INCREASING.
  Induced current creates B UP inside loop (opposing the increase).
  By right-hand rule: current goes COUNTERCLOCKWISE (viewed from above).

EXAMPLE: motional EMF on rod sliding right with B INTO page.
  Force on + charge in rod (qv × B): if v→ right, B into page,
  F = qv × B points UP along rod.
  Conventional current flows UP through rod, around loop...
  EMF = BLv,   I = BLv/R.
```

Inductors [4]/[5]

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Self-inductance:    EMF_L = −L · (dI/dt)
  L = inductance (Henry, H)

Solenoid inductance:    L = μ₀ n² · V = μ₀ N² A / L  (V = volume)

ENERGY STORED in inductor:    U_L = ½ L I²
ENERGY DENSITY of B field:     u = B²/(2μ₀)
```

RL Circuits [4]/[5]

```
CURRENT GROWING (battery V₀, R, L, switch closes):
  KVL:  V₀ − IR − L(dI/dt) = 0
  SOLUTION:  I(t) = (V₀/R)(1 − e^(−t/τ))    τ = L/R

CURRENT DECAYING (after battery removed):
  SOLUTION:  I(t) = I₀ e^(−t/τ)

Inductor INITIALLY: acts like OPEN (I = 0).
Inductor STEADY-STATE: acts like WIRE (V_L = 0).
```

LC Oscillation [5]

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Capacitor + inductor in loop (no resistor): energy oscillates between cap (E field) and inductor (B field).

GOVERNING EQUATION:    L(d²Q/dt²) + Q/C = 0
SOLUTION:    Q(t) = Q₀ cos(ωt + ϕ)        ω = 1/√(LC)
  Period:  T = 2π√(LC)

ENERGY: U_total = ½Q²/C + ½LI² = constant
Like a mass on spring (Q ↔ x, L ↔ m, 1/C ↔ k)
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KEY INSIGHT **[5] LC ↔ mass-spring analogy.** Q corresponds to position x. $I = dQ/dt$ corresponds to velocity. L plays the role of mass (inertia). $1/C$ plays the role of spring constant. $\omega = 1/\sqrt{LC}$ parallels $\omega = \sqrt{k/m}$. This unifies E&M with mechanics.

PAGE 7 | MASTER INTEGRAL PATTERNS — every C-E&M FRQ recipe

This page is your secret weapon. Like the Mechanics master patterns, every C-E&M FRQ falls into a small set of integration recipes. Recognize → set up → integrate → done.

PATTERN 1: E from continuous charge

E from line/disk/ring/rod:

1. $dq = \lambda dl$ (or σdA, or ρdV)
2. Identify symmetry — what component survives?
3. $dE_{(component)} = (k \cdot dq/r^2) \cdot \cos\theta$
4. Express r and $\cos\theta$ in terms of integration variable.
5. Integrate.

PATTERN 2: V from continuous charge

$V = \int (k \cdot dq) / r$ [SCALAR — much easier than E!]

After getting $V(r)$ or $V(z)$: $E = -dV/dr$ (or $-dV/dz$).

Often easier than direct E integration when geometry allows.

PATTERN 3: Gauss's Law application

$\oint E \cdot dA = Q_{enc} / \varepsilon_0$

Spherical: Gaussian sphere → $4\pi r^2 \cdot E = Q_{enc}/\varepsilon_0$
Cylindrical: Gaussian cylinder → $2\pi rL \cdot E = Q_{enc}/\varepsilon_0$
Planar: Gaussian pillbox → $2A \cdot E = Q_{enc}/\varepsilon_0$ (E exits both sides)

PATTERN 4: Ampère's Law application

$\oint B \cdot dl = \mu_0 I_{enc}$

Long wire (cylinder): $2\pi r \cdot B = \mu_0 I_{enc}$
Solenoid (rectangular path): N segments contribute → $BL = \mu_0 nIL$
Toroid: $2\pi r \cdot B = \mu_0 N I$

PATTERN 5: Faraday for changing flux

$EMF = -d\Phi/dt$

Setup: $\Phi = \int B \cdot dA$
Then differentiate Φ with respect to time.
Sign tells you direction (Lenz).

EXAMPLES:
Bar moving on rails: $\Phi = BLx$, $d\Phi/dt = BLv$ → $EMF = BLv$
Rotating loop: $\Phi = BA\cos(\omega t)$, $EMF = BA\omega \sin(\omega t)$

PATTERN 6: RC / RL ODE setup

RC charging: $R(dQ/dt) + Q/C = V_0$ → $Q(t) = CV_0(1 - e^{-t/RC})$
RC discharging: $R(dQ/dt) + Q/C = 0$ → $Q(t) = Q_0 e^{-t/RC}$

RL growth: $L(dI/dt) + IR = V_0$ → $I(t) = (V_0/R)(1 - e^{-Rt/L})$
RL decay: $L(dI/dt) + IR = 0$ → $I(t) = I_0 e^{-Rt/L}$

LC oscillation: $L(d^2Q/dt^2) + Q/C = 0$ → $Q(t) = Q_0\cos(t/\sqrt{LC})$

PATTERN 7: Energy density integration

E field energy density: $u_E = \frac{1}{2}\varepsilon_0 E^2$
B field energy density: $u_B = B^2/(2\mu_0)$

TOTAL ENERGY in volume V: $U = \int u \, dV$

Used to verify: $U_{cap} = \frac{1}{2}CV^2$ (integrate u_E inside capacitor)
$U_{inductor} = \frac{1}{2}LI^2$ (integrate u_B inside solenoid)

PAGE 8 | FRQ STRATEGY · MAXWELL'S EQUATIONS · SYMMETRY

Maxwell's Equations [5] — the unifying principles

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GAUSS (E):      ∮ E·dA = Q_enc/ε₀           (charges produce E)
GAUSS (B):      ∮ B·dA = 0                   (no magnetic monopoles)
FARADAY:        ∮ E·dl = −dΦ_B/dt            (changing B → E)
AMPÈRE-MAXWELL: ∮ B·dl = μ₀(I + ε₀dΦ_E/dt) (currents + changing E → B)

FROM THESE: speed of light c = 1/√(ε₀μ₀)  (links electromagnetism to optics).

AP only requires the basic Ampère + Faraday — no displacement-current calc — but
knowing the unified picture earns sophistication-level reasoning.
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Choosing Symmetry Surfaces / Loops [4]/[5]

Geometry	Surface / loop choice
Spherical (point charge, sphere of charge)	Gaussian SPHERE concentric with center → E radial, $A = 4\pi r^2$
Cylindrical (long wire, long cylinder)	Gaussian CYLINDER coaxial → E radial outward, $A_{side} = 2\pi rL$
Planar (infinite plane, parallel plates)	Gaussian PILLBOX through plane → E perpendicular, exits both ends
Long current (wire, cylindrical conductor)	Amperian LOOP circular around wire → B tangent, $\oint = 2\pi rB$
Solenoid	Amperian RECTANGLE: one side inside (BL), other 3 contribute 0

FRQ Strategy [3]/[4]/[5]

- **[3] Always start with a clear DIAGRAM** showing field/current/symmetry. Often worth 1-2 points alone.
- **[3] State the LAW you're invoking by name** before plugging in. "Applying Gauss's law to a cylindrical Gaussian surface..."
- **[4] Carry symbols, plug numbers last.** Symbolic answer is graded; numerical might still earn points if symbols correct.
- **[4] Use limit cases**: "As $r \to \infty$, $E \to 0$ ✓" or "At $z = 0$, this reduces to E_{center} ✓". Catches errors + earns sophistication.
- **[5] On RC/RL ODE problems**, always identify τ first. Then write $Q(t)$ or $I(t)$ using the standard exponential form.

Common C-E&M Traps [3]/[4]/[5]

- **[3] V is scalar, E is vector.** V from multiple charges = simple sum. E = vector sum (decompose by component).
- **[3] Inside a SOLID sphere of charge: $E \propto r$ (linear, not $1/r^2$).** Common MCQ trap.
- **[4] Magnetic force does NO work.** Only direction changes; speed stays constant.
- **[4] Capacitor combination rules OPPOSITE of resistors.** Series adds reciprocally; parallel adds directly.
- **[5] Battery connected vs disconnected with dielectric**: very different. Connected → V const, Q ↑. Disconnected → Q const, V ↓.
- **[5] Lenz's law direction**: opposes the CHANGE, not the field. If flux is decreasing, induced current REINFORCES the original direction.

PAGE 9 | MASTER EQUATION SHEET — by topic

Electrostatics

$F = kq_1q_2/r^2\ \hat{r}$ $\quad$ $E = F/q$
E from continuous: $E = \int k\cdot dq/r^2$ (vector, by component)
$V = \int k\cdot dq/r$ (scalar)
$E = -dV/dx$ (or $-\nabla V$ in 3-D)
$\oint E\cdot dA = Q_{enc}/\varepsilon_0$ (Gauss)
Potential energy: $U = qV$ $\quad$ $U = kq_1q_2/r$ (point charges)
Work: $W = q\Delta V$

Conductors + Capacitors

Conductor: $E_{inside} = 0$, V = const, charge on surface, $E_{outside} = \sigma/\varepsilon_0$
$Q = CV$ $\quad$ $C = \varepsilon_0 A/d$ (parallel plate)
$C \to \kappa C$ with dielectric
$U_{cap} = \frac{1}{2}CV^2 = \frac{1}{2}Q^2/C = \frac{1}{2}QV$
Energy density: $u_E = \frac{1}{2}\varepsilon_0 E^2$
Series: $1/C_{eq} = \Sigma(1/C_i)$ $\quad$ Parallel: $C_{eq} = \Sigma C_i$

Circuits

$V = IR$, $P = IV = I^2R = V^2/R$
Series resistors: $R_{eq} = \Sigma R$. Parallel: $1/R_{eq} = \Sigma(1/R)$
KCL: $\Sigma I_{in} = \Sigma I_{out}$. KVL: $\Sigma\Delta V = 0$

RC charging: $Q(t) = CV_0(1 - e^{-t/\tau})$, $\tau = RC$
RC discharging: $Q(t) = Q_0\, e^{-t/\tau}$
Cap initially → wire; long t → open

Magnetism + Induction

$F = qv\times B = qvB \sin\theta$ $\quad$ $F_{wire} = IL\times B$
$r = mv/(qB)$ $\quad$ $T = 2\pi m/(qB)$

Biot-Savart: $dB = (\mu_0/4\pi)(I\, dl \times \hat{r})/r^2$
$\oint B\cdot dl = \mu_0 I_{enc}$ (Ampère)
Wire: $B = \mu_0 I/(2\pi r)$. Solenoid: $B = \mu_0 nI$

$\Phi_B = \int B\cdot dA$ $\quad$ $EMF = -d\Phi/dt$ (Faraday)
L: $EMF_L = -L(dI/dt)$ $\quad$ $U_L = \frac{1}{2}LI^2$ $\quad$ $u_B = B^2/(2\mu_0)$

RL growth: $I(t) = (V/R)(1 - e^{-t\cdot R/L})$, $\tau = L/R$
LC: $\omega = 1/\sqrt{LC}$, $T = 2\pi\sqrt{LC}$

PAGE 10 | SCORE LADDER · TIER CHECKLISTS · THE ONE THING

THE SCORE LADDER

For a 3 (Pass)	For a 4	For a 5
MCQ: **16-20 / 35**	MCQ: **22-26 / 35**	MCQ: **28-32 / 35**
Coulomb's law, V = kQ/r, basic Gauss for sphere/cylinder.	Add: integration over distributions, Ampère's law, Biot-Savart on axis.	Mastery: V from distribution, then E = -dV/dx; full Maxwell perspective.
Simple circuits, RC time constant.	Multi-loop Kirchhoff, RC ODE solution, energy in cap.	Full RL + LC analysis with ODEs; energy oscillation.
Faraday for simple flux changes.	Motional EMF, sliding bar, rotating loop.	Lenz's law direction with full justification; LC ↔ mass-spring.

TIER-SPECIFIC TEST-DAY CHECKLISTS

For a 3 — focus only on these:
- ✓ MCQ: 16+ correct.
- ✓ Coulomb's law and $E = kQ/r^2$ for point charges.
- ✓ Gauss's law for spherical/cylindrical symmetry: pull E out, solve.
- ✓ Capacitors: Q = CV, $U = \frac{1}{2}CV^2$, series/parallel rules.
- ✓ Kirchhoff's laws for simple circuits.
- ✓ F = qv × B; right-hand rule.

For a 4 — add these on top:
- ✓ MCQ: 22+ correct.
- ✓ Set up E or V integral over linear/surface charge distribution.
- ✓ Apply Gauss's law for ALL three symmetries (spherical, cylindrical, planar).
- ✓ Apply Ampère's law for wires + solenoids.
- ✓ RC/RL: write ODE, identify τ, write exponential solution.
- ✓ Faraday for moving bar (BLv) and rotating loop.
- ✓ Initial-state vs steady-state analysis (cap as wire, then open).

For a 5 — add these on top:
- ✓ MCQ: 28+ correct.
- ✓ Compute V from charge distribution, then E = -dV/dx (faster than direct E).
- ✓ Energy density integration: $u_E = \frac{1}{2}\varepsilon_0 E^2$, $u_B = B^2/(2\mu_0)$.
- ✓ LC oscillation: $\omega = 1/\sqrt{LC}$, $Q(t) = Q_0 \cos(\omega t)$.
- ✓ Battery connected vs disconnected with dielectric — different answers.
- ✓ Limit-case checks on every symbolic answer.
- ✓ Maxwell's equations as unifying principles; EM waves $c = 1/\sqrt{\varepsilon_0\mu_0}$.

THE ONE THING THAT MATTERS MOST [3][4][5]: USE SYMMETRY TO PICK GAUSS OR AMPÈRE. AP Physics C: E&M is decided on whether you recognize the symmetry pattern. Spherical charge → Gauss with sphere. Cylindrical (long wire/cable) → Gauss or Ampère with cylinder. Planar (parallel plates, infinite plane) → Gauss with pillbox. Solenoid → Ampère with rectangle. **Train yourself to ask, on every FRQ involving fields: "What's the symmetry?" → then pick the right surface or loop.** Skipping this step is the #1 reason students miss FRQs they could otherwise solve.

If you do nothing else: master the 7 integral patterns on Page 7 + the Gauss/Ampère symmetry decisions, and check every answer against limit cases.

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